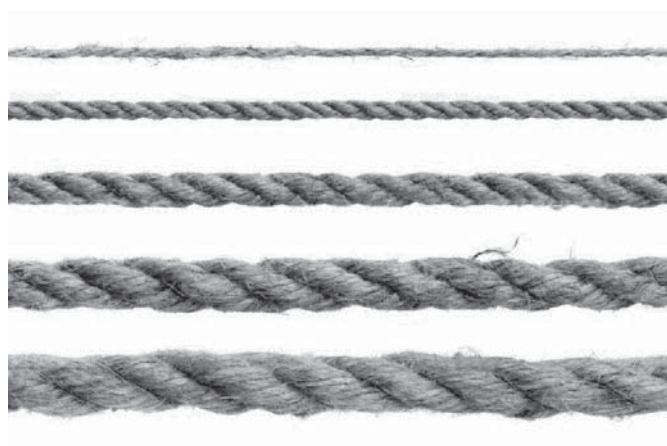


7 Some plant fibres have a high tensile strength.

The photograph shows ropes made using fibres from the stems and leaves of hemp plants.



Sclerenchyma fibres and xylem vessels are found in plant fibres.

(a) Both sclerenchyma fibres and xylem vessels contain lignin.

(i) How many of the following statements are correct?

- Lignin adds strength to the cell membranes.
- Lignin gives waterproofing properties.
- Lignin can form spiral shapes.

(1)

- ☐ A none
- ☐ B one
- ☐ C two
- ☐ D three



- (ii) Give **one** similarity, other than the presence of lignin, and **one** difference in the structures of sclerenchyma fibres and xylem vessels.

(2)

Similarity

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Difference

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- (b) Scientists have suggested that plant-based plastics should be used instead of oil-based plastics.

Hemp has been used to make some of these plant-based plastics.

Explain why hemp fibres are a sustainable resource.

(2)

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- (c) Biofuels, produced from crop plants, are a sustainable resource.

Biofuels include bioethanol and biodiesel.

Bioethanol is produced from hemp and from sugar beet. Biodiesel is produced from rapeseed oil and from soybean oil.

- (i) The table shows information about the global production of bioethanol.

Year	Global production of bioethanol $\times 10^6 \text{ dm}^3$
2000	17 000
2011	84 600

Determine what the global bioethanol production will be in 2020.

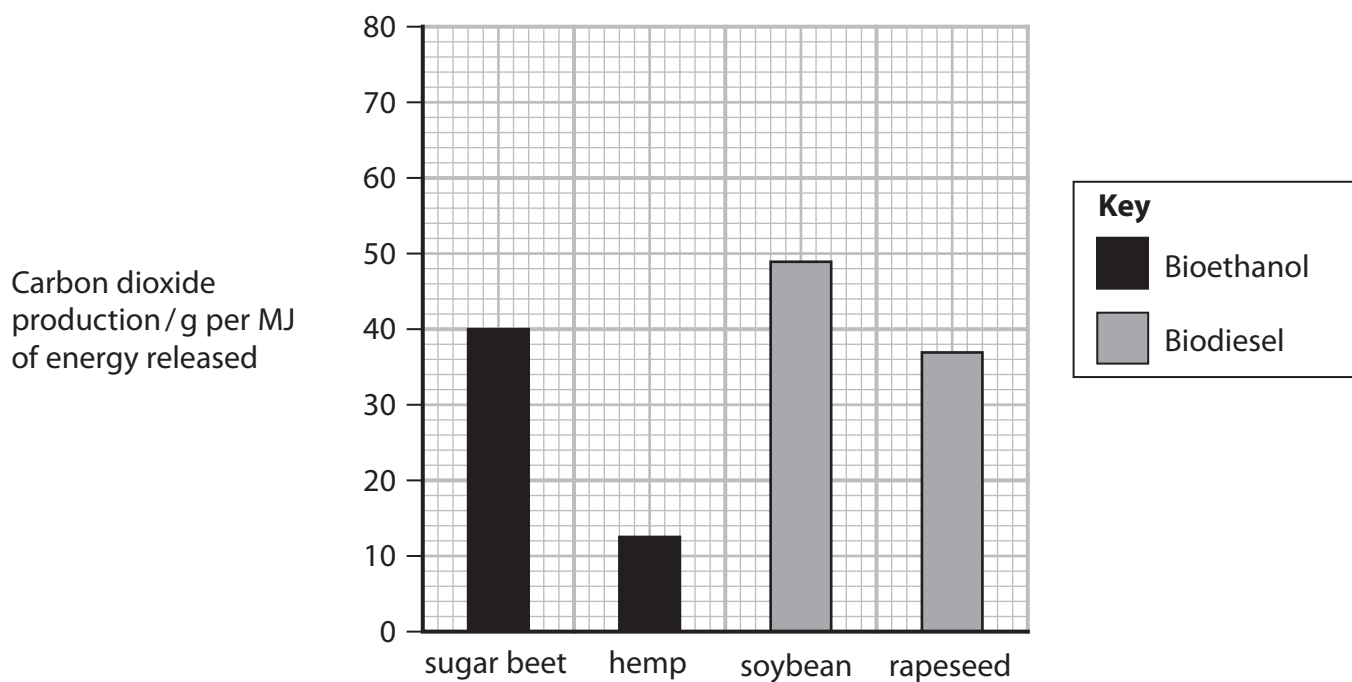
Assume the mean rate of increase stays constant.

(3)

Answer



- *(ii) The use of biofuels produces carbon dioxide. The graph shows the production of carbon dioxide using biofuels from four crop plants.



The table gives information about the level of water, fertilisers and pesticides needed for crop plants to grow.

Crop plant	Level of resources used in production of biofuels		
	Water	Fertilisers	Pesticides
Hemp	Medium to low	Low	Low
Rapeseed	High	Medium	Medium
Soybean	High	Low to medium	Medium
Sugar beet	Medium to low	High	Medium

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Explain the advantages and disadvantages of using these crop plants as a sustainable resource to produce biofuels.

Use the information in the question to support your answer.

(6)

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(Total for Question 7 = 14 marks)



P 6 2 7 9 3 A 0 2 3 2 8